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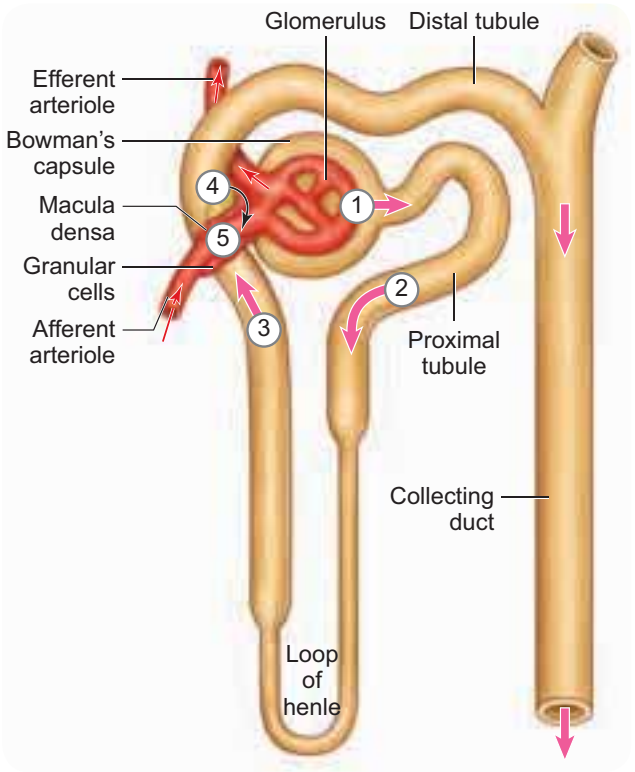


Figure 8.4 Structure of a Nephrons

[PCT] followed by a U-shaped loop of Henle (Henle's loop) that has a thin descending and a thick ascending limb. The ascending limb continues as a highly coiled tubular region called the distal convoluted tubule [DCT]. The DCT of many nephrons open into a straight tube called collecting duct. The collecting duct runs through the medullary pyramids in the region of the pelvis. Several collecting ducts fuse to form papillary duct that delivers urine into the calyces, which opens into the renal pelvis.

In the renal tubules, PCT and DCT of the nephron are situated in the cortical region of the kidney whereas the loop of Henle is in the medullary region. In majority of nephrons, the loop of Henle is too short and extends

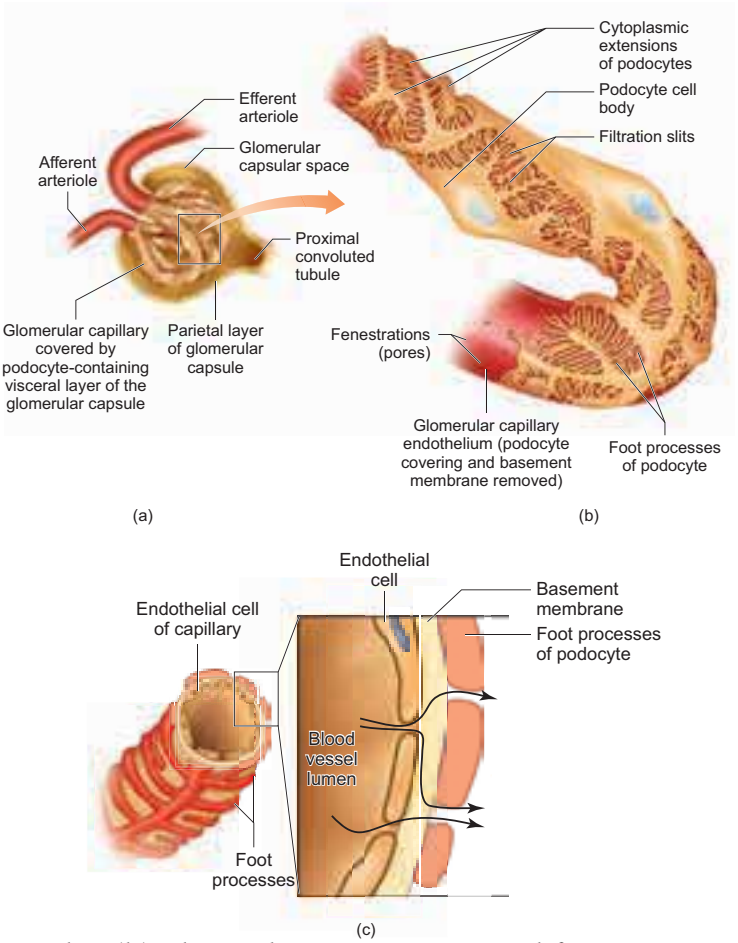


Figure 8.5 (a) Glomerulus (b) The podocytes gives several foot processes that form filtration slits (c) interacts with the basement membrane to create a filter that retains blood cells and large protein in the plasma while permitting the passage of fluids through the filtration slit.